LOT PAGER ��� 03-MECHANICAL

LOT® PAGER — Mechanical Design

1. Body

Two stainless parts, **316L** (non-magnetic, skin-safe, takes mirror polish):

- **Back plate** — flat, 40 × 40 mm, 0.8 mm. Polished (Ra ≤ 0.05 µm, #8 mirror). Nothing on it but laser-etched LOT® mark (optional, outside polished area).
- **Front shell** — 40 × 40 mm, tray form, corner radius R3, wall 0.6 mm, with cut-outs: display window, camera window, button.

Joint: laser-welded seam (sealed, IP54 target) **or** 4 × M1.2 hidden screws + silicone gasket (serviceable; recommended for P3/P5). Decide at EVT.

2. Dimensions

	v0 (build this)	v1 (stretch)	Original brief
X × Y	40 × 40 mm	40 × 40	40 × 40
Z	**9 mm**	6.5 mm	5 mm

Why 5 mm fails: see 01 §4. Camera module height alone is ≥5 mm.

Front layout (looking at side B, 40 × 40): - Display: 1.3″ window ~ 25 × 25 mm, upper-centre. - Camera: Ø3.5 mm window, top-right corner, 4 mm from edges, sapphire or Gorilla-type glass. - Button: lower-centre, 8 mm round, 0.3 mm travel dome through a thin steel membrane (or plastic insert). - Ambient-light pinhole next to camera.

3. Polishing

CNC → bead blast → mechanical polish → electro-polish. Spec Ra on the drawing. Polish is the most expensive and the most variable step: order a few extra back plates. Fingerprints show; this is the product look.

4. Manufacturing route

Route	1–10	100	Notes
CNC 316L (PCBWay CNC, or any shop)	✓	~$25–45 / set	Best for EVT/DVT
MIM 316L + polish	×	✓ (tooling $3–8k)	Consider only for 1000+

Route	1–10	100	Notes
Sheet metal + deep draw	×	×	Wall too thin for mirror finish

5. Wireless charging through steel

Qi does not work through ≥0.3 mm of stainless. Options (pick at EVT): 1. **Non-metal charging window** on side B (plastic/ceramic ring around the screen) with the coil behind it — device charges screen-down on the pad. *Recommended for v0.* 2. Ferrite-backed coil under an ultra-thin (≤0.2 mm) titanium/non-magnetic face — poor efficiency and heating. 3. Magnetic pogo pins (loses “wireless”).

The brief says one side is polished steel; with option 1 that side (A) stays steel and the device charges face-down on the pad with side B against it. Note this in 06 as the “charging orientation”.

6. RF through steel

A fully metal body is a Faraday cage. Plan: antenna on a plastic or ceramic window strip along one edge (≥ 8 × 3 mm) with u.FL → FPC antenna; tune on the real enclosure. Do not rely on the PCB antenna.

7. Thermal

ESP32-S3 + display + radio ≈ 0.5–0.9 W peak. Steel body is a good spreader; keep Wi-Fi duty < 5 % and ≤ 40 °C case.